

SEAN MACBRIDE

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RESEARCH INTERESTS

- Experimental and Observational Cosmology
- Multi-Messenger Astrophysics
- Astronomical Instrumentation Development
- Systems Design for Physics & Astronomy

EDUCATION

Ph.D. in Physics, University of Zürich May 2024 - 2027

Thesis title: *Standard Siren Cosmology: From Instrumentation to Inference*

Advisor: Professor Marcelle Soares-Santos

M.Sc. in Physics, University of Michigan-Ann Arbor Aug 2022 - May 2024

Advisor: Professor Marcelle Soares-Santos

B.A. in Physics with Honors, Wheaton College, Massachusetts Aug 2016 - May 2020

Thesis title: *Characterizing the Dust and Cold-Gas Content of Nearby Star-Forming Galaxies.*

Advisor: Professor Dipankar Maitra

SCIENTIFIC COLLABORATIONS

Rubin Observatory Legacy Survey of Space and Time (LSST) 2023 - Present

I hold roles in both the Rubin Project and the LSST Dark Energy Science Collaboration (DESC).

Rubin Observatory Project Team: Led LSST-Camera (LSSTCam) focal plane commissioning to Rubin Observatory on Cerro-Pachón in Chile, from receipt on summit through early science operations. Led testing of the focal plane in-laboratory and on-telescope. Characterized defects in the LSSTCam focal plane CCDs, and developed custom image processing tool to monitor the stability of defects in the LSSTCam focal plane. Led the Target-of-Opportunity (ToO) observing program during Rubin early operations and first year of the LSST.

LSST DESC: Forecasted the constraining power of dark sirens using simulated LSST data under different conditions. Leading a dark-siren analysis using LSST commissioning data (Data-Preview 2) and public data from the International Gravitational Wave Network (IGWN), the first dark siren measurement using LSST data.

International Gravitational-Wave Observatory Network (IGWN) 2025 - Present

Led dark siren analysis using DECam Local Volume Exploration survey (DELVE) survey data for IGWN O4c data, including galaxy catalog preparation and systematic-uncertainty quantification. Coordinated time-critical ToO observations with IGWN researchers, interfacing with Rubin Observatory alert brokers to rapidly disseminate information about gravitational wave (GW) candidates. Contributed to dark siren analysis using Dark Energy Survey (DES) data for IGWN O4b data.

Dark Energy Survey (DES) 2023 - Present

Performed observations, image processing, and analysis of images from Dark Energy Camera (DECam) observations in response to GW triggers from the IGWN observing run 4 (O4) for the Dark Energy Survey gravitational wave (DESGW) observing program. Implemented improvements to DESGW alert and image processing pipelines, including improvements to the telescope strategy and GW alert processing. Led collaboration between the Japanese collaboration for Gravitational wave ElectroMagnetic follow-up (J-GEM) and DESGW through joint observations of GW event S250328ae using DECam and the Prime Focus Spectrograph (PFS) at Subaru Observatory.

Tested Dark Energy Spectroscopic Instrument (DESI) robotic positioners to increase movement precision and improve low performing robotic positioners on the DESI focal plane. Characterized a common failure mode of DESI positioners and resolved the failure with a reproducible solution. Authored and submitted an internal report and presented the results of this study to the DESI focal plane working group. Led dynamic testing of prototype robotic positioner performance through lifetime, focal-ratio degradation, and fiber angle tests using different telescope configurations. Tested prototype robotic positioners for Stage-V surveys, focusing on characterization of tilt induced by different orientations.

Large Lenslet Array Magellan Spectrograph (LLAMAS)

2020 - 2022

Assembled the opto-mechanical mounts for the LLAMAS instrument spectrograph. Led verification and testing of mechanical tolerance for custom-fabricated parts with a coordinate measurement machine. Verified spectrograph grating efficiency through optical tests, and integrated the spectrograph camera lenses into their bezels for the 24 camera spectrograph. Integrated the 2,400 AR-coated fibers to the spectrograph by bonding with optical adhesive, with zero fiber loss. LLAMAS was integrated to the Magellan Baade telescope, and achieved first light in November 2024.

PROFESSIONAL APPOINTMENTS, SERVICE, AND LEADERSHIP

2026 - 2028	LSST DESC, Speakers Bureau Co-Chair
2026 - 2028	LSST DESC, Collaboration Council Member
2025 - 2027	Rubin Observatory, ToO Observer
2024 -	Rubin Observatory, LSSTCam Summit Support Scientist
2022 - 2024	University of Michigan Physics Graduate Council, Elected member
2022 - 2024	University of Michigan Physics DEI committee, Member
2022 - 2024	University of Michigan APS chapter, Member
2016 - 2020	Wheaton College physics club, Member

APPROVED OBSERVING PROGRAMS

2025	GW-Primo. PI: McMahon, Co-I: MacBride. 2 nights on the Blanco telescope at CTIO.
2023 - 2027	DESGW. PI: Soares-Santos, Co-I: MacBride. 4 nights on the Blanco telescope at CTIO, every semester (6 months).

SELECTED PUBLICATIONS

Below are selected publications from different collaborations and projects. A full list of my publications can be found on my Google Scholar profile or on NASA ADS. My h-index is 8 as of September 5, 2026.

Astronomical Instrumentation

1. The On-Sky Performance of the LSST Camera CCD Array. **MacBride**, Roodman, ... 30 Jun 2026.
2. First year into operations of LLAMAS, the Large Lenslet Array Magellan Spectrograph: design, build and commissioning overview, results and performance, lessons learned. Furesz, ... **MacBride**, ... 21 Aug 2026.
3. Characterizing robotic positioners under the influence of changing gravity vectors for future spectroscopic surveys. Wüthrich, ... **MacBride**, ... 25 Jun 2026.

Observational and Experimental Cosmology

1. Measurement of the Hubble constant using the Dark Energy Survey Year 6 Gold galaxy catalog and the fourth Gravitational-Wave Transient Catalog. McMahon, ... **MacBride**, ... 5 Feb 2026.
2. Standard Sirens in 2040s: Probing the Cosmic Expansion History with Gravitational Waves and Spectroscopic Galaxy Surveys. Borghi, ... **MacBride**, ... 20 Dec 2025.

Multi-Messenger Astrophysics

1. A Joint Search for the Electromagnetic Counterpart to the Gravitational-Wave Binary Black-Hole Merger Candidate S250328ae with the Dark Energy Camera and the Prime Focus Spectrograph. Zhang, Kokubo, **MacBride**, ... 1 Aug 2025.
2. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1) . Chandler, Bernardelli, ... **MacBride**, ... 21 July 2025.
3. Rubin ToO 2024: Envisioning the Vera C. Rubin Observatory LSST Target of Opportunity program. Andreoni, Margutti, ... **MacBride** ... 7 Nov 2024.

Astronomical Systems Development

1. The Rubin Observatory Target-of-Opportunity Program in the First Year of Operations. **MacBride**, Ribeiro, ... 30 Jun 2026.
2. LSSTCam installation and commissioning. Fanning, ... **MacBride**, ... 21 Aug 2026.
3. Scheduling the LSST: commissioning the Rubin Feature Based Scheduler. Jones, ... **MacBride**, ... 19 Aug 2026.

RESEARCH GRANTS

2024 - 2026 *Quantifying the Impact of Galaxy Catalog Conditions for Dark Sirens*, 3M CPU hour grant National Energy Research Scientific Computing Center, ScienceAtScale

INVITED TALKS

09/2026	Colloquium, Transients and Variable Stars Science Collaboration Colloquium, Virtual
07/2026	Talk, Rubin Community Workshop 2026, SLAC National Accelerator Laboratory, Menlo Park, CA, USA
07/2026	Plenary, Rubin Community Workshop 2026, SLAC National Accelerator Laboratory, Menlo Park, CA, USA
07/2026	Seminar, Dark Energy School, LSST DESC Collaboration Meeting, Paris, France
07/2026	Talk, SPIE Astronomical Telescopes + Instrumentation, Copenhagen, Denmark
05/2026	Talk, Rubin ToO 2026 Workshop, Virtual
11/2025	Talk, Wheaton College, Norton, MA, USA
11/2025	Talk, Rubin-LSST France Meeting at LPSC Grenoble, France
09/2025	Talk, 5th Philip Wetton Workshop, Oxford University, Oxford, Great Britain
07/2025	Talk, Rubin Community Workshop 2025, Tucson, AZ, USA
06/2025	Colloquium, Cal-Tech LIGO Seminar Series, Virtual
06/2025	Seminar, Massachusetts Institute of Technology, Cambridge, MA, USA
02/2025	Plenary, LSST DESC Collaboration meeting, Virtual

CONTRIBUTED TALKS

08/2026	Talk, Rubin 2036, SLAC National Accelerator Laboratory, Menlo Park, CA, USA
07/2026	Talk, Rubin Community Workshop 2026, SLAC National Accelerator Laboratory, Menlo Park, CA, USA
07/2026	Talk, Rubin Community Workshop 2026, SLAC National Accelerator Laboratory, Menlo Park, CA, USA
07/2026	Talk, SPIE Astronomical Telescopes + Instrumentation, Copenhagen, Denmark
04/2026	Seminar, Stanford University, Stanford, CA, USA
03/2026	Talk, LIGO-Virgo-KAGRA Collaboration Meeting, Pisa, Italy
03/2026	Talk, LIGO-Virgo-KAGRA Collaboration Meeting, Pisa, Italy
05/2025	Talk, The Dark Energy Survey Collaboration Meeting 2025, Virtual
03/2025	Talk, APS March meeting 2025, Anaheim CA, USA
05/2020	Talk, Wheaton College, MA, USA
09/2018	Seminar, Wheaton College, Norton MA, USA
08/2018	Talk, Rutgers University, New Brunswick, NJ, USA

CONFERENCE ORGANIZATION

2025	EAS Annual Meeting Special Session 27: The Rubin Observatory Legacy Survey of Space and Time: from Data Previews to Full Survey Operations, co-chair, local organizing committee
2024	Conferences for Undergraduate Women in Physics (CUWiP) 2024, local organizing committee
2023	DESI Collaboration Meeting, local organizing committee
2023	University of Michigan-Ann Arbor Physics Graduate Student Symposium, chair local organizing committee

TEACHING

2026	Introduction to Astroparticle Physics (PHY473), University of Zürich, Graduate Instructor
2026	Introduction to Scientific Computing (PHY124), University of Zürich, Graduate Instructor
2025	Introduction to Astroparticle Physics (PHY473), University of Zürich, Graduate Instructor
2023 - 2024	Introductory Physics for Life Sciences (PHYSICS 151/251), University of Michigan-Ann Arbor, Lead Graduate Instructor
2022 - 2023	Introductory Physics for Life Sciences (PHYSICS 151/251), University of Michigan-Ann Arbor, Graduate Instructor
2018 - 2019	Introductory Physics (PHYS 170/171/180/181), Wheaton College, MA, Undergraduate Teaching Assistant and Tutor

MENTORSHIP

2026	Alexs Gaute: Summer Masters Student at University of Zürich.
2025 - 2026	Guandi Zhao: Masters Student at University of Zürich.
2024 - 2025	Banan Yamani: Masters Student at University of Zürich. Current Physics PhD student at Paul Scherrer Institute.
2024 - 2025	Yavuz Gencel: Masters Student at University of Zürich.
2023 - 2024	Elise Fuller: Undergraduate student at University of Michigan-Ann Arbor. Current physics PhD student at CU Boulder.
2023 - 2024	Tim Fanning: Undergraduate student at University of Michigan-Ann Arbor. Current controls engineer specialist at Argonne National Laboratory.

OUTREACH

2026	Led ‘Ask a Scientist’ sessions focused on observational cosmology and gravitational waves for the Science Pavilion at UZH, as part of the City of Zurich’s Long Night of Museums, an annual event where Zurich’s museums invite visitors to engage with science, culture, and art museums free of charge.
2025	Led tour groups and interviews with media members at Rubin Observatory, focusing on LSSTCam. Visting parties included including media members, friends and family visits, tour groups, visiting project review teams from the National Science Foundation and Department of Energy, and political dignitaries.
2023 - 2024	Led group of high school students in Burlington VT to observe the 2023 annular eclipse and 2024 total eclipse. Prepared and delivered lectures for students about the physics of the sun, solar observations, and eclipses through virtual and in person presentations.
2023	Designed art installation in University of Michigan Physics help room, highlighting struggles overcome by recent graduates of the physics program and success in their later lives.
2016 - 2020	Showcased features of different telescopes to local tour groups at Wheaton College Observatory. Operate and adjust telescopes to show appropriate stars, objects, and events based on sky conditions. Educated children and adults in how to locate objects and explained features about the objects on sky.

IN PRESS

08/2025	The Vera Rubin Observatory is ready to revolutionize astronomy - Science News.
06/2025	Mapping the Universe at the Rubin Observatory in Chile - BBC.
06/2025	World’s Largest Digital Camera Snaps Its First Photos of the Universe - Wall Street Journal.
04/2024	NASA volunteer documents eclipse from Burlington to help study sun’s atmosphere - Vermont Public Radio.
04/2024	Vermont native and NASA-sponsored researcher returns to Burlington to study solar eclipse - NBC5.